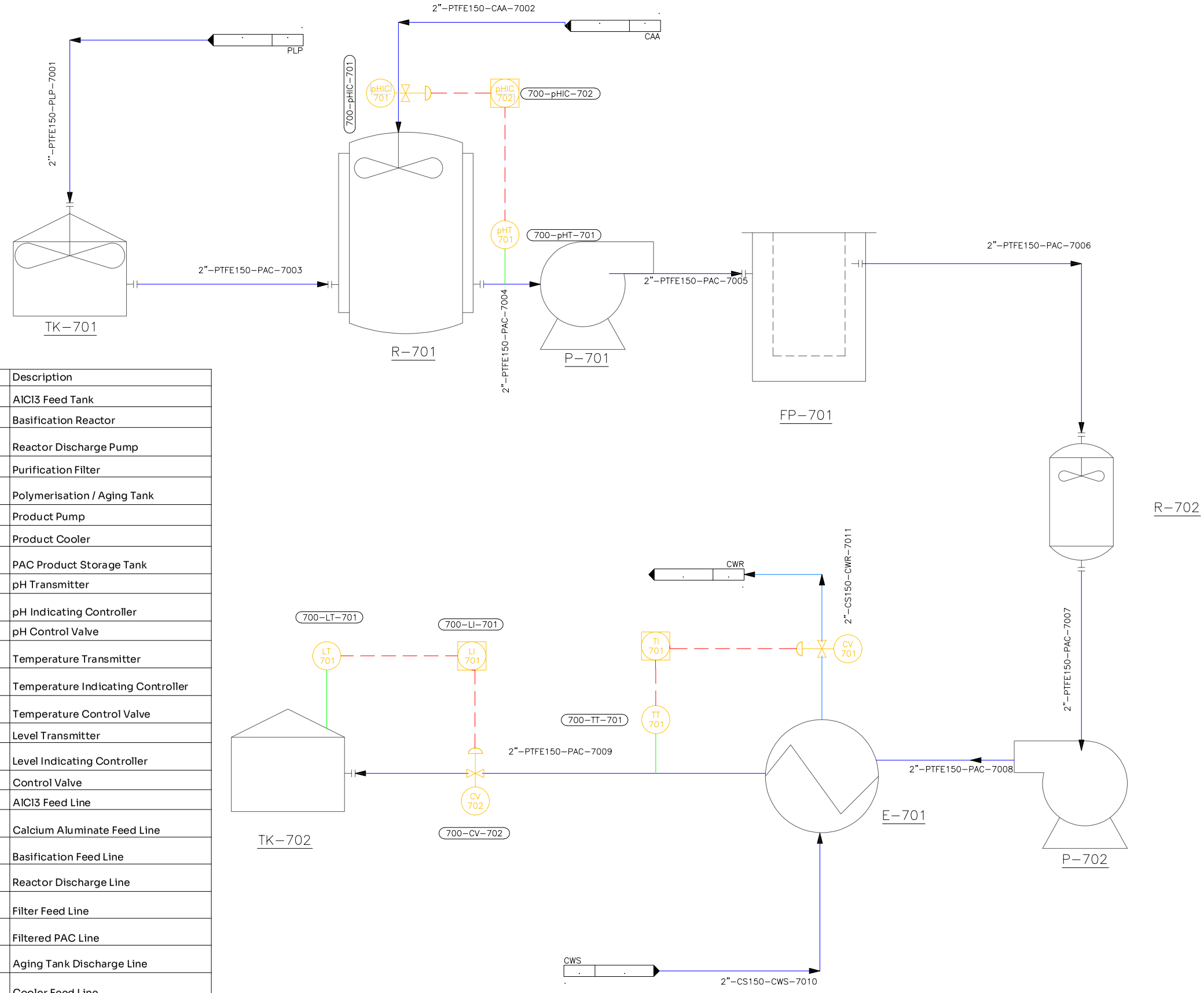


Tag Number	Category	Description
TK-701	Mechanical Equipment	AlCl3 Feed Tank
R-701	Mechanical Equipment	Basification Reactor
P-701	Mechanical Equipment	Reactor Discharge Pump
FP-701	Mechanical Equipment	Purification Filter
R-702	Mechanical Equipment	Polymerisation / Aging Tank
P-702	Mechanical Equipment	Product Pump
E-701	Mechanical Equipment	Product Cooler
TK-702	Mechanical Equipment	PAC Product Storage Tank
pHT-701	Instrumentation	pH Transmitter
pHIC-701	Instrumentation	pH Indicating Controller
pHCV-701	Instrumentation	pH Control Valve
TT-701	Instrumentation	Temperature Transmitter
TIC-701	Instrumentation	Temperature Indicating Controller
TV-701	Instrumentation	Temperature Control Valve
LT-701	Instrumentation	Level Transmitter
LIC-701	Instrumentation	Level Indicating Controller
CV-702	Instrumentation	Control Valve
2"-PTFE150-PLP-7001	Piping	AlCl3 Feed Line
2"-PTFE150-CAA-7002	Piping	Calcium Aluminate Feed Line
2"-PTFE150-PAC-7003	Piping	Basification Feed Line
2"-PTFE150-PAC-7004	Piping	Reactor Discharge Line
2"-PTFE150-PAC-7005	Piping	Filter Feed Line
2"-PTFE150-PAC-7006	Piping	Filtered PAC Line
2"-PTFE150-PAC-7007	Piping	Aging Tank Discharge Line
2"-PTFE150-PAC-7008	Piping	Cooler Feed Line
2"-PTFE150-PAC-7009	Piping	Cooled PAC Product Line
2"-CS150-CWS-7010	Piping	Cooling Water Supply Line
2"-CS150-CWR-7011	Piping	Cooling Water Return Line

General Notes

1. All pipes, fittings, and valves shall be designed in accordance with ASME B31.3 process piping standards.
2. All instrumentation symbols and nomenclature conform to ISA 5.1 standards.
3. Fail–safe actions for all pneumatic control valves (fail–open/fail–closed) shall be determined during the HAZOP review phase.
4. Due to the highly corrosive nature of Aluminum Chloride and Polyaluminum Chloride (PAC), all process piping (designated as PTFE150) and wetted equipment surfaces must be strictly PTFE–lined, FRP, or manufactured from chemically compatible materials.
5. The basification reaction in R–701 relies heavily on accurate pH control. The pH probe requires a regular calibration and maintenance schedule to prevent over–dosing or under–dosing of Calcium Aluminate (CAA).
6. PTFE–lined pipes have strict upper temperature limits. The Product Cooler must maintain the final PAC product temperature strictly within the safe operating margins of the storage tank and downstream piping.
7. All transfer and product pumps handling corrosive slurries and PAC solutions require specialized mechanical seals (e.g., double mechanical seals with barrier fluid) to prevent acid leaks.

No.	Revision/Issue	Date

Drawing Name

Area700

Basification, Polymerisation, and Purification

Project Name and Address

PAC_P&ID

Final Year Project

PMB, UPO, KNUST, Ghana

Kumasi Ashanti Region

Area	Stamp
700	
Drawing Number: 7	
Author: GROUP 7B	

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